

1. Problem Statement

Problem

Training institutes that offer AI and Agentic AI programs face several challenges in reaching the right audience.

Currently,

- Finding engineering colleges across India is mostly manual.
- Identifying the correct departments (CSE, AI&ML, Data Science, IoT, ECE, etc.) takes significant effort.
- Collecting Training & Placement Officer (TPO), Innovation Cell, Incubation Cell, and Faculty Coordinator contacts is time-consuming.
- Marketing campaigns are generic and not personalized.
- Follow-up with colleges is inconsistent.
- Lead tracking is done manually using spreadsheets.
- There is no centralized system to manage outreach, campaigns, and analytics.

Solution

Develop an Agentic AI Marketing Platform that automates the complete college outreach process using multiple AI agents.

The system will:

- Identify target colleges
- Detect eligible departments
- Find official faculty contacts
- Generate personalized marketing content
- Launch email/WhatsApp/LinkedIn campaigns
- Track responses
- Qualify leads
- Schedule follow-ups
- Maintain CRM
- Provide real-time analytics

2. Who are the End Users?

Primary Target Audience

- Educational Institutions
- Engineering Colleges
- Autonomous Institutions
- Universities
- Degree Colleges offering Computer Science programs
- Departments
- Computer Science Engineering (CSE)
- CSE (Artificial Intelligence & Machine Learning)
- CSE (Data Science)
- Computer Science
- Internet of Things (IoT)
- Electronics and Communication Engineering (ECE)
- Allied Computer Science Branches
- College Officials
- Training & Placement Officers (TPO)
- Placement Coordinators
- Innovation Cell Coordinators
- Incubation Cell Coordinators
- Heads of Departments
- Faculty Coordinators
- Principal / Dean (where applicable)
- Student Audience
- Final Year B.Tech Students
- Fresh Graduates
- Students interested in AI
- Students looking for internships
- Students preparing for placements

3. Project Features

College Discovery

Discover engineering colleges across India

Categorize by state, city, university affiliation, and institution type

Department Detection

Automatically identify relevant departments

- CSE
- AI & ML
- Data Science
- IoT
- ECE
- Allied branches

Contact Discovery

- Collect official
- Faculty coordinators
- TPO emails
- Placement cell contacts
- Innovation cell contacts
- Incubation cell contacts
- Lead Qualification

Score each college based on

- AI-related departments
- Placement activity
- Innovation ecosystem
- Student strength
- Previous collaborations
- Personalized Campaigns

Generate personalized

- Emails
- WhatsApp messages
- LinkedIn messages
- Brochures
- Landing pages
- CRM Management
- Lead status
- Campaign history
- Meeting schedules
- Notes
- Follow-up reminders
- Campaign Automation
- Multi-channel campaigns
- Email automation
- Reminder scheduling
- Follow-up automation
- Analytics Dashboard

Track

- Emails sent
- Open rate
- Click rate
- Replies
- Qualified leads
- Conversion rate
- Colleges onboarded

4. Which Agents are Required?

The architecture consists of 18 specialized AI agents.

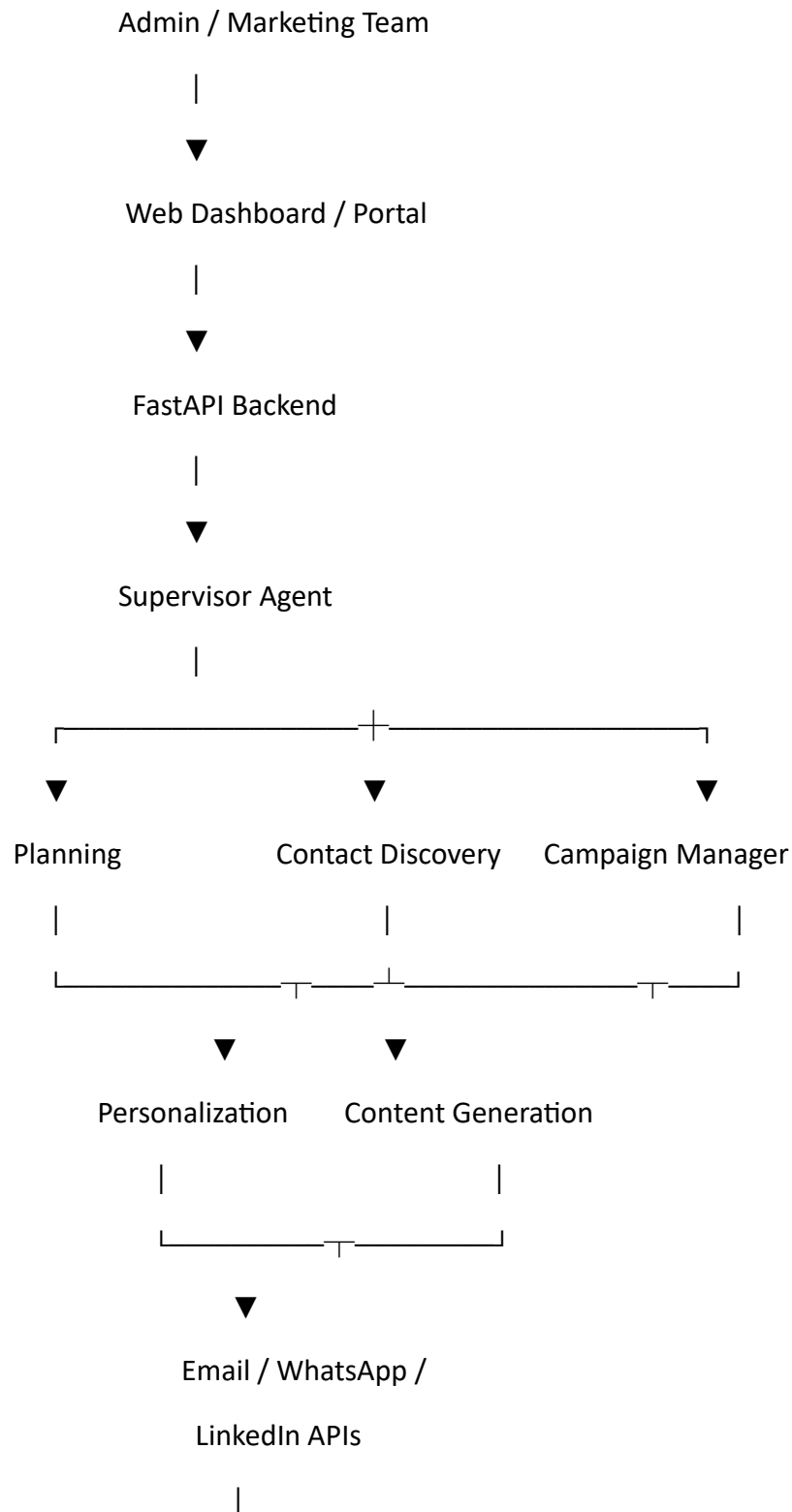
- Agent Responsibility
- Supervisor Agent-Coordinates all agents and workflow
- Website Creation Agent-Creates landing pages and campaign websites
- Planning Agent-Detects departments and plans outreach strategy
- Routing Agent-Performs opportunity analysis and routes tasks
- Memory Agent-Stores campaign history and discovered contacts
- Contact Discovery Agent-Finds official college contacts
- Policy Agent-Performs lead qualification based on predefined rules
- Monitoring Agent-Monitors campaign execution and agent health
- Content Generation Agent-Generates emails, brochures, and marketing content
- College Personalization Agent-Personalizes communication for each college
- Campaign Manager Agent-Executes and manages campaigns
- CRM Agent-Maintains lead pipeline and interaction history
- Follow-up Agent-Sends automated reminders and follow-ups
- Analytics Agent-Generates reports and campaign insights
- Email Agent-Sends personalized emails
- Schedule Agent-Books meetings and manages calendars
- Database Agent-Manages data storage and retrieval
- Reporting Agent-Generates daily, weekly, and monthly reports

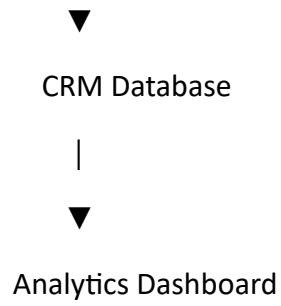
5. End-to-End Workflow

1. START
2. Target Engineering Colleges
3. College Discovery
4. Department Detection
5. Faculty & TPO Contact Discovery
6. Lead Qualification
7. Opportunity Analysis
8. College Personalization
9. AI Content Generation
10. Campaign Creation
11. Multi-Channel Campaign Execution
12. (Email • WhatsApp • LinkedIn)
13. CRM Update
14. Lead Pipeline Tracking
15. Automated Follow-up Scheduling
16. Meeting Booking & Calendar Integration
17. Campaign Analytics & Performance Monitoring
18. Reports & Interactive Dashboard

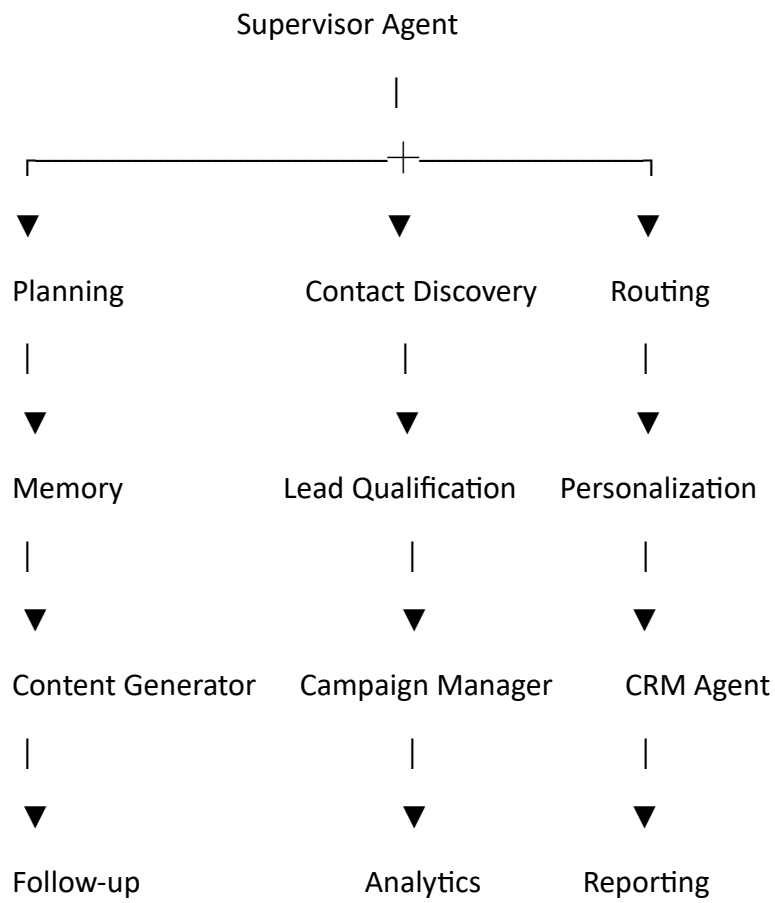
6. System Architecture

6.1 High-Level Architecture





6.2 Agent Architecture



6.3 Database Architecture

- PostgreSQL
- Colleges
- Departments
- Faculty
- Leads
- Campaigns
- CRM
- Users
- Meetings
- Reports
- MongoDB
- Chat history
- Agent logs
- Email templates
- Campaign responses
- Vector Database (ChromaDB / FAISS)
- College information
- Marketing documents
- Prompt library
- Email templates
- Previous campaign knowledge

6.4 API Architecture

- Internal APIs
- College Search API
- Department Detection API
- Contact Discovery API
- Lead Qualification API
- Campaign API
- CRM API
- Analytics API

- Reporting API
- External APIs
- Google Maps / Places
- Google Search
- LinkedIn
- Email APIs (SMTP, Gmail, SendGrid)
- Calendar APIs
- LLM APIs (OpenAI, Gemini, Claude)

6.5 Suggested Project Folder Structure

agentic-ai-marketing/

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|— agents/

| |— supervisor.py

| |— planning_agent.py

| |— routing_agent.py

| |— memory_agent.py

| |— contact_discovery.py

| |— policy_agent.py

| |— monitoring_agent.py

| |— personalization_agent.py

| |— content_generation.py

| |— campaign_manager.py

| |— crm_agent.py

| |— followup_agent.py

| |— analytics_agent.py

| └─ reporting_agent.py

|

|─ prompts/

|─ tools/

|─ models/

|─ apis/

|─ database/

|─ workflows/

|─ campaigns/

|─ analytics/

|─ tests/

|─ config/

|─ crm/

|─ main.py

└─ requirements.txt